

2 Methods

2.1 Data and Variables

Only scripts `02_setup_and_global_variables.R` and `03_biomarker_strategies.R` were sourced. No survival-model fitting scripts were run.

The available repository extract contains 74 patients after scripts 02 and 03. The main complete-case dataset used for non-TLR survival and interaction models contains 68 patients. TLR in the DAG corresponds to `tlr` in the data, and the TLR subset with observed `tlr` contains 68 patients.

Patients without follow-up CT assessment have `tlr = NA` and were excluded from TLR analyses. Those analyses therefore rely on a missing-at-random assumption conditional on the observed covariates included in each model.

Table 1: Analysis datasets used in the DAG association report

Dataset	N	Deaths	Progressions
Source dataset after scripts 02 and 03	74	65	52
Main complete-case dataset for survival and interaction models	68	59	49
TLR subset with observed <code>tlr</code>	68	59	50

For the interaction node, T was coded as `T_num = 1` for the experimental arm and 0 for control, and `TxTMB` was defined as `T_num * tmb_braf`.

2.2 Statistical Tests

Each observable edge was tested using the prespecified method from the plan:

- Wilcoxon rank-sum for continuous-versus-binary comparisons
- Fisher’s exact test for 2 x 2 tables
- Mantel-Haenszel tests for binary associations conditional on `tmb_braf`
- Firth-corrected Cox regression (`coxphf`) for time-to-event endpoints
- Firth-corrected logistic regression (`logistf`) for TLR models
- A conditional permutation test (`coin::independence_test`) for `Age || TxTMB | {TMB_BRAF}`

The two arrows into `TxTMB` (`T -> TxTMB` and `TMB_BRAF -> TxTMB`) are definitional because `TxTMB` is a constructed interaction node. The four edges from the latent node `U` are not empirically testable and are noted only in prose.

2.3 Multiple Testing

No multiplicity correction was applied in this version of the report. All statistical summaries therefore use raw p-values only, and the findings should be interpreted as exploratory rather than confirmatory.

3 Direct Edge Tests

3.1 Covariate edges (`Age/Sex -> TMB_BRAF`)

Age and sex are the only baseline covariates with direct arrows into `TMB_BRAF`. The age association was tested with a Wilcoxon rank-sum test and the sex association with Fisher’s exact test.

3.2 Latent edges (U -> *)

The DAG includes four latent edges: U -> TMB_BRAF, U -> CRP, U -> PFS, and U -> OS. Because U is unobserved, these edges are not testable in the trial data and are therefore not assigned p-values.

3.3 Biomarker edges (TMB_BRAF, CRP -> PFS, OS, TLR)

Direct biomarker effects on PFS and OS were tested with univariable Firth Cox models in `data_main`. Direct effects on TLR were tested in the `tlr` subset using Firth logistic regression.

3.4 Treatment edges (T -> PFS, OS, TLR)

Treatment effects were evaluated with `T_num` as a binary predictor, preserving the randomised-arm coding from the trial extract.

3.5 Interaction edges (TxTMB -> PFS, OS)

The TxTMB node was tested using the interaction coefficient from `Surv(...) ~ T_num + tmb_braf + T_num:tmb_braf` for both PFS and OS. These are the same interaction-scale questions emphasized in `clinical_effectiveness.qmd`, but used here as DAG edge checks rather than biomarker-model comparisons.

3.6 Mediator edge (TLR -> PFS)

The TLR -> PFS edge was estimated in patients with observed `tlr`. This is a subset analysis because TLR is undefined when no follow-up scan is available.

3.7 Outcome chain (PFS -> OS)

The PFS -> OS edge was represented by a Firth Cox model for OS with `PFSwk` entered as a continuous predictor.

3.8 Edge summary table

Table 2: Direct edge tests

Edge	Test	N	Effect	p-value	Conclusion
Age -> TMB_BRAF	Wilcoxon rank-sum	69	Median Age: TMB/BRAF=0=61.0, TMB/BRAF=1=65.0	0.450	No association detected (p >= 0.05)
Age -> OS	Firth Cox	68	1.00 (95% CI 0.98 to 1.03)	0.861	No association detected (p >= 0.05)
Sex -> TMB_BRAF	Fisher's exact	69	0.87 (95% CI 0.30 to 2.48)	0.812	No association detected (p >= 0.05)
TMB_BRAF -> PFS	Firth Cox	68	0.40 (95% CI 0.21 to 0.75)	0.004	Association detected (p < 0.05)
TMB_BRAF -> OS	Firth Cox	68	0.68 (95% CI 0.40 to 1.13)	0.139	No association detected (p >= 0.05)
TMB_BRAF -> TLR	Firth logistic	65	0.92 (95% CI 0.34 to 2.52)	0.877	No association detected (p >= 0.05)
CRP -> PFS	Firth Cox	68	0.40 (95% CI 0.20 to 0.75)	0.004	Association detected (p < 0.05)
CRP -> OS	Firth Cox	68	0.49 (95% CI 0.27 to 0.84)	0.009	Association detected (p < 0.05)
CRP -> TLR	Firth logistic	68	3.48 (95% CI 1.14 to 12.59)	0.028	Association detected (p < 0.05)
T -> PFS	Firth Cox	68	0.81 (95% CI 0.45 to 1.45)	0.465	No association detected (p >= 0.05)
T -> OS	Firth Cox	68	0.96 (95% CI 0.58 to 1.61)	0.873	No association detected (p >= 0.05)
T -> TLR	Firth logistic	68	0.33 (95% CI 0.11 to 0.90)	0.030	Association detected (p < 0.05)
TLR -> PFS	Firth Cox (TLR subset)	68	0.18 (95% CI 0.09 to 0.33)	<0.001	Association detected (p < 0.05)
PFS -> OS	Firth Cox	68	0.98 (95% CI 0.97 to 0.99)	<0.001	Association detected (p < 0.05)
TxTMB -> PFS	Firth Cox interaction	68	0.37 (95% CI 0.11 to 1.24)	0.106	No association detected (p >= 0.05)
TxTMB -> OS	Firth Cox interaction	68	0.63 (95% CI 0.22 to 1.79)	0.385	No association detected (p >= 0.05)

4 Conditional Independence Tests

4.1 Randomisation balance (marginal CIs involving T)

The four CI statements involving T double as randomisation checks. If any of them were significant, that would suggest baseline imbalance in this realized sample rather than a structural failure of the DAG.

Table 3: Randomization balance checks implied by the DAG

Item	p-value	Conclusion
Age _ _ T	0.172	Compatible with randomization balance
CRP _ _ T	0.023	Possible randomization imbalance
Sex _ _ T	0.254	Compatible with randomization balance
T _ _ TMB_BRAF	1.000	Compatible with randomization balance

One randomization-balance check, CRP _||_ T, was significant on the raw scale ($p = 0.023$). The other three randomization checks remained non-significant.

4.2 Structural CIs (conditional)

The DAG implies 13 observable conditional independencies after excluding the latent node U. These were tested using the prespecified conditional models or stratified tests.

4.3 CI summary table

Table 4: Conditional-independence tests

CI statement	Test	N	Effect	p-value	Randomization check	Conclusion
Age _ _ CRP	Wilcoxon rank-sum	71	Median Age: CRP=0=65.0, CRP=1=64.5	0.961	No	Compatible with DAG-implied
Age _ _ Sex	Wilcoxon rank-sum	74	Median Age: Sex=0=65.5, Sex=1=64.5	0.961	No	Compatible with DAG-implied
Age _ _ T	Wilcoxon rank-sum (randomization check)	74	Median Age: T=0=65.5, T=1=60.5	0.172	Yes	Compatible with randomization
Age _ _ TLR {CRP, TMB_BRAF}	Firth logistic	65	1.03 (95% CI 0.98 to 1.08)	0.292	No	Compatible with DAG-implied
Age _ _ TxTMB {TMB_BRAF}	Conditional permutation test	69	Standardized Z = 0.77	0.457	No	Compatible with DAG-implied
CRP _ _ Sex	Fisher's exact	71	0.74 (95% CI 0.25 to 2.23)	0.619	No	Compatible with DAG-implied
CRP _ _ T	Fisher's exact (randomization check)	71	3.51 (95% CI 1.12 to 12.10)	0.023	Yes	Possible randomization imbalance
CRP _ _ TxTMB {TMB_BRAF}	Mantel-Haenszel	68	21.67 (95% CI 2.23 to 210.11)	0.007	No	CI contradicted by data
Sex _ _ T	Fisher's exact (randomization check)	74	0.58 (95% CI 0.21 to 1.59)	0.254	Yes	Compatible with randomization
Sex _ _ TxTMB {TMB_BRAF}	Mantel-Haenszel	69	0.52 (95% CI 0.12 to 2.17)	0.592	No	Compatible with DAG-implied
Sex _ _ TLR {CRP, TMB_BRAF}	Firth logistic	65	1.13 (95% CI 0.40 to 3.21)	0.812	No	Compatible with DAG-implied
TLR _ _ TxTMB {T, TMB_BRAF}	Firth logistic	65	2.31 (95% CI 0.29 to 19.27)	0.426	No	Compatible with DAG-implied
T _ _ TMB_BRAF	Fisher's exact (randomization check)	69	0.96 (95% CI 0.33 to 2.76)	1.000	Yes	Compatible with randomization

5 Consolidated Summary Table

Table 5: Consolidated summary of direct edge and conditional-independence tests

	Domain	Item	Effect	p-value	Conclusion
∞	Direct edge	Age -> TMB_BRAF	Median Age: TMB/BRAF=0=61.0, TMB/BRAF=1=65.0	0.450	No association detected (p >= 0.05)
	Direct edge	Age -> OS	1.00 (95% CI 0.98 to 1.03)	0.861	No association detected (p >= 0.05)
	Direct edge	Sex -> TMB_BRAF	0.87 (95% CI 0.30 to 2.48)	0.812	No association detected (p >= 0.05)
	Direct edge	TMB_BRAF -> PFS	0.40 (95% CI 0.21 to 0.75)	0.004	Association detected (p < 0.05)
	Direct edge	TMB_BRAF -> OS	0.68 (95% CI 0.40 to 1.13)	0.139	No association detected (p >= 0.05)
	Direct edge	TMB_BRAF -> TLR	0.92 (95% CI 0.34 to 2.52)	0.877	No association detected (p >= 0.05)
	Direct edge	CRP -> PFS	0.40 (95% CI 0.20 to 0.75)	0.004	Association detected (p < 0.05)
	Direct edge	CRP -> OS	0.49 (95% CI 0.27 to 0.84)	0.009	Association detected (p < 0.05)
	Direct edge	CRP -> TLR	3.48 (95% CI 1.14 to 12.59)	0.028	Association detected (p < 0.05)
	Direct edge	T -> PFS	0.81 (95% CI 0.45 to 1.45)	0.465	No association detected (p >= 0.05)
	Direct edge	T -> OS	0.96 (95% CI 0.58 to 1.61)	0.873	No association detected (p >= 0.05)
	Direct edge	T -> TLR	0.33 (95% CI 0.11 to 0.90)	0.030	Association detected (p < 0.05)
	Direct edge	TLR -> PFS	0.18 (95% CI 0.09 to 0.33)	<0.001	Association detected (p < 0.05)
	Direct edge	PFS -> OS	0.98 (95% CI 0.97 to 0.99)	<0.001	Association detected (p < 0.05)
	Direct edge	TxTMB -> PFS	0.37 (95% CI 0.11 to 1.24)	0.106	No association detected (p >= 0.05)
	Direct edge	TxTMB -> OS	0.63 (95% CI 0.22 to 1.79)	0.385	No association detected (p >= 0.05)
	Conditional independence	Age _ _ CRP	Median Age: CRP=0=65.0, CRP=1=64.5	0.961	Compatible with DAG-implied CI
	Conditional independence	Age _ _ Sex	Median Age: Sex=0=65.5, Sex=1=64.5	0.961	Compatible with DAG-implied CI
	Conditional independence	Age _ _ T	Median Age: T=0=65.5, T=1=60.5	0.172	Compatible with randomization balance
	Conditional independence	Age _ _ TLR {CRP, TMB_BRAF}	1.03 (95% CI 0.98 to 1.08)	0.292	Compatible with DAG-implied CI
	Conditional independence	Age _ _ TxTMB {TMB_BRAF}	Standardized Z = 0.77	0.457	Compatible with DAG-implied CI
	Conditional independence	CRP _ _ Sex	0.74 (95% CI 0.25 to 2.23)	0.619	Compatible with DAG-implied CI
	Conditional independence	CRP _ _ T	3.51 (95% CI 1.12 to 12.10)	0.023	Possible randomization imbalance
	Conditional independence	CRP _ _ TxTMB {TMB_BRAF}	21.67 (95% CI 2.23 to 210.11)	0.007	CI contradicted by data
	Conditional independence	Sex _ _ T	0.58 (95% CI 0.21 to 1.59)	0.254	Compatible with randomization balance
	Conditional independence	Sex _ _ TxTMB {TMB_BRAF}	0.52 (95% CI 0.12 to 2.17)	0.592	Compatible with DAG-implied CI
	Conditional independence	Sex _ _ TLR {CRP, TMB_BRAF}	1.13 (95% CI 0.40 to 3.21)	0.812	Compatible with DAG-implied CI
	Conditional independence	TLR _ _ TxTMB {T, TMB_BRAF}	2.31 (95% CI 0.29 to 19.27)	0.426	Compatible with DAG-implied CI
	Conditional independence	T _ _ TMB_BRAF	0.96 (95% CI 0.33 to 2.76)	1.000	Compatible with randomization balance

6 Discussion

The direct-edge analysis found 7 of 16 tested edges with $p < 0.05$. Direct edges with $p < 0.05$: TMB_BRAF $\rightarrow$ PFS; CRP $\rightarrow$ PFS; CRP $\rightarrow$ OS; CRP $\rightarrow$ TLR; T $\rightarrow$ TLR; TLR $\rightarrow$ PFS; PFS $\rightarrow$ OS. The strongest supported pattern sits along the prognostic and disease-course portion of the DAG: low CRP was associated with lower hazards for both PFS and OS, TMB_BRAF was associated with lower PFS hazard, TLR was strongly associated with subsequent PFS, and longer PFS was tightly linked to longer OS. In other words, the data support the idea that baseline biomarkers and intermediate disease response are more informative than age or sex for the downstream survival relationships represented here.

Treatment-related edges were still mixed. T $\rightarrow$ PFS, T $\rightarrow$ OS, and both TxTMB interaction terms remained non-significant, so the report still does not provide strong evidence for treatment effect modification through the TMB/BRAF interaction node. However, the raw-p analysis now shows $p < 0.05$ for both CRP $\rightarrow$ TLR ($p = 0.028$) and T $\rightarrow$ TLR ($p = 0.030$), which suggests that tumour lesion reduction may be one route through which baseline inflammation and treatment assignment connect to later outcomes.

Among the 13 conditional-independence tests, 11 were compatible with the DAG at $p \geq 0.05$ and 2 were contradicted by the data. Conditional independencies contradicted by the data at $p < 0.05$: CRP $\parallel$ T; CRP $\parallel$ TxTMB $\mid$ {TMB_BRAF}. In practice, both violated statements involve CRP: CRP $\parallel$ T had $p = 0.023$, suggesting possible baseline imbalance by treatment arm, and CRP $\parallel$ TxTMB $\mid$ {TMB_BRAF} had $p = 0.007$, suggesting residual dependence around the CRP-treatment-interaction part of the graph. That makes CRP the main place where the current DAG looks least secure in this dataset.

Interpretation should remain cautious. The latent node U is unobserved, TLR analyses depend on non-missing follow-up imaging, the available repository extract is modest in size, and several tests are necessarily low-power. These checks therefore speak to empirical compatibility, not causal proof.

7 Conclusion

This report adds a data-facing complement to the causal DAG by testing its observable edges and implied conditional independencies in the METIMMOX-1 repository extract. Overall, the empirical signal is stronger for prognostic and disease-course relationships than for treatment-effect-modification relationships: CRP, TMB_BRAF, TLR, and PFS show the clearest links to later outcomes, while the direct treatment and TxTMB interaction edges remain uncertain in this small sample.

Taken together, the observed data support the core survival pathway (TLR $\rightarrow$ PFS $\rightarrow$ OS) and several biomarker-survival links, but they also surface two CRP-related tensions with the assumed graph. The most defensible takeaway is therefore that the DAG captures much of the prognostic structure in the data, while the CRP-related balance and interaction structure may need refinement or stronger justification in future versions.

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